Davis CA Intel: A Source-Bound Regional Intelligence Deployment

Version 0.1.0 — review draft; not a published release

Daniel Ari Friedman

2026-10-31

Abstract

Davis CA Intel is a configured inland deployment of the Region Intel civic intelligence package. This software case study describes how a thin regional repository can retain its own identity, provider policy and public evidence while importing shared acquisition, validation, search and publication methods. The recorded Davis municipal edition contains 290 article pages and 2541 sections from eCode360 DA4922. Its verification receipt is dated 2026-10-31T03:39:57.381Z. Those counts describe the accepted source snapshot; they are not estimates of legal completeness or current civic activity. We present a reproducible manuscript pipeline that binds authored prose and claim evidence to exact public input bytes, then produces native PDF and HTML review editions. The configured catalog contains 10 source declarations and enables 8 inland monitors. Configuration counts are distinguished from successful collection. No civic-impact experiment, human usability study, model-answer evaluation or independent legal review is reported. The contribution is a source-bound deployment and review workflow, with explicit limits on what its software and artifact checks establish.

Introduction

Local civic information is distributed across municipal codes, agency notices, meeting references and time-sensitive feeds. Combining these materials creates two distinct problems: implementing reusable data-handling methods and deciding which sources, geographic policies and interpretations belong to a locality. A coastal application's defaults are especially unsuitable as implicit policy for an inland city. A second deployment therefore provides a useful engineering case for examining the boundary between shared computation and local evidence.

This paper concerns Davis, California, in Yolo County. It describes the public Davis repository and its dependency on Region Intel. The original Crescent City application is a compatibility preset in that package; its manuscript findings and coastal source coverage are not Davis observations. The case study asks whether deployment identity, accepted corpus evidence and manuscript numbers can remain independently attributable while implementation is shared.

The design draws on the distinction between provenance entities, activities and responsible agents in PROV-DM (Moreau and Missier 2013). Input hashes and execution receipts are operational commitments, not a claim that this implementation is a complete PROV serializer. FAIR provides a relevant vocabulary for reusable metadata and identifiable inputs (Wilkinson et al. 2016); this paper does not claim a formal FAIR assessment or unrestricted redistribution rights. The following sections describe a reproducible engineering record, rather than an assessment of governmental performance or community outcomes.

Methods

Deployment admission and isolation

The consumer's versioned deployment document binds a civic profile, regional provider policy, active output directory, public seed directory, enabled roster and source catalog. The loader validates profile-policy identity before running producers. The civic profile names eCode360 DA4922, the Davis corpus/vector namespace and a distinct calendar identity. The monitoring anchor and watch bounds are operating policy, not surveyed city boundaries. Generic methods and algorithm tests remain in Region Intel; the consumer contains configuration, its importer, regional evidence and deployment acceptance.

Municipal acquisition and acceptance

Acquisition uses a candidate root separate from active output and public seeds. The accepted edition must have complete expected article/section membership, local source/extraction custody and exact manifest, TOC and article-set bindings. The recorded verification also includes a current TOC comparison and a bounded deterministic live sample. A passing sample is not an instantaneous check of every upstream page. A source-marked reserved or repealed section may retain an empty body only when the source heading and parsed status agree. Unmarked empty extraction is rejected. Failure or interruption leaves the candidate unaccepted; bounded, journaled promotion preserves an earlier complete active edition.

Manuscript derivation

The manuscript reads the accepted public seed, not whichever live producer output happens to be newest. The shared builder validates the seed's v2 verification and exact publication-input receipt, recomputes article and section denominators from the public code projection, and captures configuration, dependency lock, bibliography, claim ledger and numbered source bytes. It substitutes declared evidence tokens and records an input fingerprint. This follows the reproducibility principle that a reported result should retain its computational inputs and procedure (Sandve et al. 2013). Rendering is a later operation with tool versions and output hashes; it does not update source clocks.

Evidence scope

The unit of analysis is one accepted Davis edition and its configured deployment. Article pages are provider acquisition units; sections are retained municipal records, including explicitly inactive entries. Catalog entries and enabled adapters are configuration units. They are not interchangeable denominators, and none is a count of local events, citizens, legal obligations or hazards.

Results

Recorded Davis edition

The accepted public projection contains 290 article pages and 2541 sections. The recorded verification time is 2026-10-31T03:39:57.381Z. The verification report declares passing local, current-TOC and sample planes, with no missing sections in its expected denominator. These statements report the retained receipt and public projection; preparing this paper did not perform a new live acquisition or independently interpret the law.

Recorded item	Value	Meaning
Municipality	DA4922	Provider identity, not a geography survey
Article pages	290	Accepted provider acquisition units
Sections	2541	Retained section records, including inactive headings
Source declarations	10	Configured catalog, not collection completeness
Enabled monitors	8	Selected adapters, not verified hazard coverage

Inland policy and public surfaces

The configured monitors cover earthquake, weather, air quality, wildfire, drought, smoke, roads and California statewide fuel observations. Marine, tide, harbor and other coastal adapters are excluded. The full policy schema still contains compatibility fields for disabled adapters; their presence is not evidence of Davis coverage. Official-page meeting collection retains bounded links and explicit unavailable states, rather than claiming a complete archive.

The static edition searches the accepted public code projection. The local dashboard's corpus search uses separately acquired and promoted raw articles. Optional retrieval-augmented generation requires its own model/vector services and accepted index. Neither a configured model nor a renderable paper supplies an evaluated Davis model-answer result. Software test counts and browser acceptance are intentionally kept in their versioned engineering receipts rather than presented here as measurements of civic usefulness.

Discussion

The Davis case makes the dependency boundary inspectable: the consumer names a single immutable shared revision and owns the configuration and evidence that give that implementation a local meaning. This arrangement reduces the need to maintain copied methods, but it does not eliminate locality-specific decisions. Provider access, geographic filters, data rights and interpretation remain deployment responsibilities. The existence of a second repository is evidence of an implemented configuration seam, not proof that every supported adapter works in every municipality.

The manuscript pipeline applies the same separation to reporting. Authored arguments are versioned locally, computed counts come from a captured seed, and rendered outputs have a distinct byte receipt. This is useful for tracing which edition a reader is examining. It is insufficient to determine whether an ordinance was adopted, whether a cited rule remains effective, or whether a particular interface supports a resident's decision. Those require different evidence and appropriately qualified review.

Further empirical work could study reader tasks, source-specific temporal coverage and independently interpreted legal examples. Such work would need declared sampling units, recruitment or review procedures, comparison criteria and retained disagreements. No result from those proposed studies is inferred from the present artifact record.

Limitations, Ethics, and Threats to Validity

Time and completeness. Local hash agreement concerns retained bytes. A live TOC and deterministic sample at acquisition time do not prove continuing all-page currency, comprehensive City communications or legal completeness. Changing upstream pages can invalidate later interpretation without changing a retained seed. The verification timestamp must remain visible.

Coverage and interpretation. Configured zones, radii, counties and road queries are explicit operating choices. Missing, unavailable, stale and empty states are distinct; an inaccessible source cannot become evidence of quiet conditions. Statewide fuel observations are not a Davis pump-price survey. Meeting links are not verified proceedings, votes or complete archived content. No independent legal review, human usability assessment or civic-impact estimate is supplied by this paper.

Models and automation. No model-generated answer is evaluated in this study. Shared model functionality cannot be promoted to Davis factuality or usefulness by reusing another region's findings. AI-assisted drafting and software editing do not constitute independent human peer review. Responsibility for final authorship, interpretation and publication declarations remains with the author.

Rights and disclosure. The repository's software/manuscript licensing does not grant additional rights over third-party source material. Preserve source attribution and applicable provider terms. Public review artifacts exclude credentials, raw acquisition HTML and private operational diagnostics. Funding, conflict and final contribution declarations require owner confirmation before publication; no declaration is inferred from missing metadata.

Reproducibility and Artifact Specification

The Davis consumer installs its full public Git dependency SHA through a frozen Bun lockfile. It does not require a sibling core checkout. From that checkout, the review workflow is:

```
bun install --frozen-lockfile
bun run validate
bun run manuscript:check
bun run manuscript:render
bun run manuscript:validate-render
```

The renderer uses native Pandoc citation processing and XeLaTeX for PDF and standalone HTML for web review. It disables raw HTML/TeX in source Markdown, uses a finite owned child for each tool, checks the PDF with qpdf and extracts text for unresolved-content checks. Exact tool versions and member hashes are recorded alongside the rendered edition. Failed builds preserve the prior complete edition; successful rendering alone does not imply peer review.

The input fingerprint is recorded in the appendix and evidence JSON. It binds the accepted seed, configuration, dependency manifest and lock, authored source, bibliography, claim ledger and renderer source. Re-checking an edition compares current inputs and exact output members. Rendered PDF timestamps may differ across independent builds; compare the declared inputs and semantic content separately from the receipt that identifies one actual PDF/HTML byte edition.

The public data and source are available at the Davis CA Intel repository. Raw source custody and acquisition diagnostics remain local; the public seed retains their commitments rather than claiming that omitted raw files can be reconstructed from a digest. This distinction limits independent re-extraction from the public bundle and should guide any replication claim.

Conclusion

Davis CA Intel demonstrates a configured regional consumer that imports shared methods while retaining its own accepted municipal evidence and manuscript sources. The recorded 290-article, 2541-section edition supports a precise description of this software artifact. It does not establish ongoing legal currency, complete source access, evaluated model factuality or public benefit. A separately bound native review edition makes those reporting limits and reproducibility inputs inspectable before a 0.1 publication decision.

Appendix: Formal Contracts and Worked Records

For this review edition, a claim-evidence entry names captured public file paths and states both its scope and what is not established. The counts in the Results table are substituted from validated seed records rather than entered by hand. The complete file-by-file byte commitments are in `evidence.json`; rendered members and tool versions are in `review-receipt.json`.

```
profile: davis-ca
municipality: DA4922
source edition:
2d74bf8205ecde96b88b4183d4d47169d83ee910e5d1ce125fd6c77bd31eca8b
shared runtime:
7cb15fa1af0455e4d756551ddf828da0cbf5095e
manuscript input fingerprint:
2d34a2664e2f140bdee02d53277b93225a76122f8848ed3620d00d74c5067af3
```

Three separate acceptance questions remain visible: whether captured inputs match their declared edition; whether the PDF/HTML render matches the captured manuscript inputs; and whether reviewers accept the claims and interpretation. The software checks the first two operational boundaries. It cannot certify the third by writing a successful receipt or a version number.

References

Bibliographic entries cited in the text follow below. The source receipt and repository files supply the empirical artifact record; the scholarly sources provide context for provenance, metadata reuse and computational reproducibility.

Moreau, Luc, and Paolo Missier. 2013. *PROV-DM: The PROV Data Model*. W3C Recommendation. World Wide Web Consortium. https://www.w3.org/TR/2013/REC-prov-dm-20130430/.

Sandve, Geir Kjetil, Anton Nekrutenko, James Taylor, and Eivind Hovig. 2013. "Ten Simple Rules for Reproducible Computational Research." *PLOS Computational Biology* 9 (10): e1003285. https://doi.org/10.1371/journal.pcbi.1003285.

Wilkinson, Mark D., Michel Dumontier, IJsbrand Jan Aalbersberg, et al. 2016. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3: 160018. https://doi.org/10.1038/sdata.2016.18.